Rahel Tesfaye

Research Data Scientist · Zurich, Switzerland · rahel@example.com

I build the software that decides whether a wiggle in a detector is a pair of black holes, a passing truck, or a fault in a cooling pump. Most of my days are spent on noise, so that the rare real signal gets an honest answer fast.

EXPERIENCE

Research Data Scientist, Limmat Institute for Wave Physics — Zurich · Mar 2023 to Present

- Lead data analysis for the institute's low-latency gravitational-wave search, from whitened strain to a published sky map.
- Built Chirpline, the four-stage pipeline that now runs during every observing period, with a median of 38 seconds from data to posterior.
- Designed the glitch taxonomy detector teams use to trace noise to its source, cutting false triggers in the main search by 41%. Supervise two doctoral students.

Data Scientist, Kestrel Seismic Analytics — Zurich · Jan 2021 to Feb 2023

- Built anomaly detection for a network of 600 seismic stations, separating real ground motion from sensor faults and traffic.
- Shipped an earthquake early warning model with a median alert lead time of 4.8 seconds on held-out events.
- Moved the team's models from nightly batch jobs to a streaming service with monitoring and automatic rollback.

Doctoral Researcher, Physics, Halden University of Technology — Halden · Sep 2016 to Dec 2020

- Thesis on neural estimators for short transient signals in non-Gaussian detector noise.
- Wrote the first version of the group's open-source glitch simulator, still used in teaching.

FLAGSHIP

Chirpline: From raw detector strain to a sky map in under a minute.

Chirpline is the low-latency pipeline I designed for the institute's gravitational-wave search. Median 38 seconds from data to posterior. Replays two years of archived data as a nightly regression test. Runs on a 12 GPU cluster during observing periods.

SELECTED PROJECTS

- Glitchbook, 2024: A self-supervised map of the short noise bursts that fake gravitational-wave signals. (31 glitch families)
- Flowcast, 2025: Amortised posteriors for compact binary signals with a conditional normalizing flow. (2.1 s per event posterior)
- Line Hunter, 2023: Tracks the narrow spectral lines that drift through detector data, so searches can subtract them. (180 lines tracked live)
- Quiet Hours, 2024: Change-point detection on environmental sensors to flag bad stretches of data. (0.94 ROC-AUC on labelled segments)
- Synthetic Noise Bench, 2025: A diffusion model that generates realistic non-Gaussian detector noise for injection studies. (9 TB synthetic noise a week)
- Ringdown Check, 2025: Neural ratio estimation to test whether a merger remnant rings with more than one tone. (5,000 simulations per calibration)
- Scatter Arches, 2023: A small CNN that finds scattered-light arches in spectrograms before they hit the search. (92% recall at 1% false alarms)
- Segment Atlas, 2024: A web tool for browsing data quality flags, glitches and lines across a whole observing period. (40 analysts each week)
- Tremor Watch, 2022: Early warning across 600 seismic stations, with sensor faults filtered out first. (4.8 s median alert lead time)
- Calibration Emulator, 2024: A Gaussian process emulator for detector calibration uncertainty, fast enough to sample in every analysis. (0.6% max amplitude error)

PUBLICATIONS AND TALKS

- Amortised Posteriors for Compact Binary Signals with Conditional Flows. JCAM 2025, Journal article, 2025.
- Thirty-One Kinds of Noise: Self-Supervised Glitch Taxonomy for Interferometer Data. ACSPP 2024, Alpine Conference on Signal Processing in Physics, Davos, 2024. Best Paper Award.
- What the Noise Knows: Anomaly Detection for Detectors That Never Sit Still. Invited talk, Zurich Data Science Colloquium, 2025.

EDUCATION

PhD, Physics, Halden University of Technology — 2020

MSc, Applied Mathematics, Halden University of Technology — 2016

SKILLS

Signal processing: Matched filtering, Whitening, Spectral estimation, Kalman filters, Wavelets, Time-frequency analysis
Machine learning: PyTorch, JAX, Self-supervised learning, Normalizing flows, Diffusion models, Anomaly detection, Scikit-learn
Inference and statistics: Simulation-based inference, Neural ratio estimation, Nested sampling, Gaussian processes, Bayesian change points, Calibration checks
Engineering: Python, TypeScript, Kafka, Kubernetes, Dask, DuckDB, Grafana, Git

CERTIFICATIONS

Advanced Bayesian Workflow, Rhine Summer School of Statistics (Aug 2022); GPU Programming for Scientists, Halden Computing Centre (Mar 2020); Research Software Engineering, Nordic Research Software Network (Jun 2019)

Sample portfolio. The person, companies and projects are fictional.